

NE 523: Homework 2

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1 Distributions & Point Functions

Suppose

$$\phi(E) = 1/E \tag{1a}$$

$$\sigma_s(E) = \sigma_0 \tag{1b}$$

$$\sigma_a(E) = \sigma_{a0} \sqrt{E_0/E} \tag{1c}$$

$$u = \ln(E_0/E) \tag{1d}$$

A Functions of Lethargy

1. We can begin by writing E in terms of u

$$\exp(u) = E_0/E \Rightarrow E = E_0 \exp(-u) \tag{2}$$

2. Substitute the above relation into the expression for $\phi(E)$

$$\phi(u) = \frac{\exp(u)}{E_0} \tag{3}$$

3. The scattering cross section does not change.

$$\sigma_s(u) = \sigma_0 \tag{4}$$

4. Substitute into $\sigma_a(E)$

$$\sigma_a(u) = \sigma_{a0} \sqrt{\exp(u)} \tag{5}$$

B Converting Eq. 1-21

1. From L&M, we have Eq. 1-21

$$f_n(E \rightarrow E', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{1}{(1-\alpha)E} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & \alpha E < E' < E \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

2. Plug in the relationship from part a.

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{\exp(u)}{(1-\alpha)E_0} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & \frac{\alpha E_0}{\exp(u)} < u' < \frac{E_0}{\exp(u)} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

C Integrate $f_n(u)$

1. First, let's split the integral into three separate integrals. The bounds of these integrals are given by the conditional evaluation of $f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}')$ in part b. I will neglect the variable dependencies of f_n on for brevity.

$$\int du' \int_{4\pi} d\hat{\Omega}' f_n = \left[\int_0^{\frac{\alpha E_0}{\exp(u)}} du' \int_{4\pi} d\hat{\Omega}' 0 \right] + \left[\int_{\frac{\alpha E_0}{\exp(u)}}^{\frac{E_0}{\exp(u)}} du' \int_{4\pi} d\hat{\Omega}' \frac{\exp(u)}{(1-\alpha)E_0} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}) \right] + \left[\int_{\frac{E_0}{\exp(u)}}^{\infty} du' \int_{4\pi} d\hat{\Omega}' 0 \right] \quad (8)$$

$$\int du' \int_{4\pi} d\hat{\Omega}' f_n = C_1 + \left[\int_{\frac{\alpha E_0}{\exp(u)}}^{\frac{E_0}{\exp(u)}} du' \int_{4\pi} d\hat{\Omega}' \frac{\exp(u)}{(1-\alpha)E_0} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}) \right] + C_2 \quad (9)$$

2. The meaning of the above integral is the integral the probability density function, f_n , over all phase space. In the regions where $f_n = 0$, we intuitively understand there will be no contribution to the total integral. Therefore, $C_1, C_2 = 0$.

$$\int du' \int_{4\pi} d\hat{\Omega}' f_n = \int_{\frac{\alpha E_0}{\exp(u)}}^{\frac{E_0}{\exp(u)}} du' \int_{4\pi} d\hat{\Omega}' \frac{\exp(u)}{(1-\alpha)E_0} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}) \quad (10)$$

3. Begin by factoring.

$$\frac{\exp(u)}{(1-\alpha)E_0} \int_{\frac{\alpha E_0}{\exp(u)}}^{\frac{E_0}{\exp(u)}} du' \int_{4\pi} d\hat{\Omega}' \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}) \quad (11)$$

4. Regardless of the dependence of $\tilde{\mu}$ on angle, we can use the definition of the delta function to integrate the integral over angle.

$$\int_{4\pi} d\hat{\Omega}' \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}) = 1 \quad (12)$$

5. Then, integrate over lethargy space.

$$\int_{\frac{\alpha E_0}{\exp(u)}}^{\frac{E_0}{\exp(u)}} du' = \left[u \right]_{\frac{\alpha E_0}{\exp(u)}}^{\frac{E_0}{\exp(u)}} = \frac{E_0}{\exp(u)} - \frac{\alpha E_0}{\exp(u)} \quad (13)$$

6. Plug both integral back into the total integral.

$$\int du' \int_{4\pi} d\hat{\Omega}' f_n = \frac{\exp(u)}{(1-\alpha)E_0} \left[\frac{E_0}{\exp(u)} - \frac{\alpha E_0}{\exp(u)} \right] \quad (14)$$

$$\int du' \int_{4\pi} d\hat{\Omega}' f_n = \frac{\exp(u)}{(1-\alpha)E_0} \frac{E_0}{\exp(u)} (1-\alpha) \quad (15)$$

7. All variables cancel and we arrive at the expected result.

$$\int du' \int_{4\pi} d\hat{\Omega}' f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = 1 \quad (16)$$

2 Solving The Transport Equation for the Uncollided Flux

1. Start with the neutron transport equation.

$$\left[\hat{\Omega} \cdot \hat{\nabla} + \Sigma_t(\vec{r}, \hat{\Omega}, E) \right] \psi(\vec{r}, \hat{\Omega}, E) = q_{ex} + \int dE' \int d\hat{\Omega}' \Sigma_s(\vec{r}, \hat{\Omega}, E) \psi(\vec{r}, \hat{\Omega}', E') \quad (17)$$

2. There is no distributed source and we are only finding the uncollided flux, so both terms on the right hand side are zero.

$$\left[\hat{\Omega} \cdot \hat{\nabla} + \Sigma_t \right] \psi_0(\vec{r}, \hat{\Omega}, E) = 0 \quad (18)$$

3. In 1D,

$$\hat{\Omega} \cdot \hat{\nabla} = \mu \frac{d}{dx} \quad (19)$$

4. The equation becomes:

$$\mu \frac{d\psi_0(x, \mu, E)}{dx} + \Sigma_t \psi_0(x, \mu, E) = 0 \quad (20)$$

$$\frac{d\psi_0(x, \mu, E)}{dx} = -\frac{\Sigma_t}{\mu} \psi_0(x, \mu, E) \quad (21)$$

5. Splitting the derivative and integrating,

$$\frac{d\psi_0(x, \mu, E)}{\psi_0(x, \mu, E)} = -\frac{\Sigma_t}{\mu} dx \quad (22)$$

$$\int \frac{d\psi_0}{\psi_0} = \int dx \left[-\frac{\Sigma_t}{\mu} \right] \quad (23)$$

$$\ln(\psi_0) = -\frac{\Sigma_t x}{\mu} + C_1 \quad (24)$$

$$\psi_0 = \exp \left(\frac{-\Sigma_t x}{\mu} + C_1 \right) = C_1 \exp \left(\frac{-\Sigma_t x}{\mu} \right) \quad (25)$$

6. We have the following two boundary conditions. The first BC is from mono-energetic neutron source incident on the left face and the second BC is from the vacuum BC on the right face. Note, the mono-energetic, fixed-direction source is represented as the product of the source intensity and two delta functions in angle and energy.

$$\psi_0(0, \mu, E) = S_{\text{left planar}} = I \delta(\mu - \tilde{\mu}) \delta(E - E_0), \quad \mu > 0 \quad (26a)$$

$$\psi_0(a, \mu, E) = 0, \quad \mu \leq 0 \quad (26b)$$

7. Using our BC from Eq. [26a](#), we can find the value of C_1 .

$$\psi_0(0, \mu, E) = I \delta(\mu - \tilde{\mu}) \delta(E - E_0) = C_1 \exp(0) \quad (27)$$

$$C_1 = I \delta(\mu - \tilde{\mu}) \delta(E - E_0) \quad (28)$$

8. We are then left with the final expression for ψ_0 .

$$\psi_0(x, \mu, E) = I \exp\left(\frac{-\Sigma_t x}{\mu}\right) \delta(\mu - \tilde{\mu}) \delta(E - E_0) \quad (29)$$

3 The Once Collided Flux

1. We start by recognizing the scattering term we omitted when calculating ψ_0 will be the source for ψ_1 .

Note, a homogeneous medium implies $\Sigma_s \neq f_n(x)$.

$$q_1 = \int dE' \int d\mu' \Sigma_s(\mu', E') \psi_0 \quad (30)$$

$$= \int dE' \int d\mu' \Sigma_s(\mu' \rightarrow \mu, E' \rightarrow E) I \exp\left(\frac{-\Sigma_t x}{\mu}\right) \delta(\mu - \tilde{\mu}) \delta(E - E_0) \quad (31)$$

2. Resulting in the final expression for q_1 :

$$q_1 = \Sigma_s(\tilde{\mu} \rightarrow \mu, E_0 \rightarrow E) I \exp\left(\frac{-\Sigma_t x}{\tilde{\mu}}\right) \quad (32)$$

3. From Problem 2, we know ψ_1 is given a similar form as ψ_0 , but with a source of q_1 .

$$\mu \frac{d\psi_1(x, \mu, E)}{dx} + \Sigma_t \psi_1(x, \mu, E) = q_1 \quad (33)$$

4. Substituting q_1 into the expression for ψ_1 :

$$\mu \frac{d\psi_1(x, \mu, E)}{dx} + \Sigma_t \psi_1(x, \mu, E) = \Sigma_s(\tilde{\mu} \rightarrow \mu, E_0 \rightarrow E) I \exp\left(\frac{-\Sigma_t x}{\tilde{\mu}}\right) \quad (34)$$

4 Solving the Integral Transport Equation

1. Start with the integral transport equation for case of isotropic scattering and isotropic source:

$$\phi(\vec{r}, E) = \int_V dV' \frac{\exp(-\tau(E; \vec{r}' \rightarrow \vec{r}))}{4\pi|\vec{r}' - \vec{r}|^2} \int_0^\infty [dE' [\Sigma_s(\vec{r}, E' \rightarrow E) + \chi(E)\nu(E')\Sigma_f(\vec{r}, E')] \phi(\vec{r}', E') + S(\vec{r}', E)] \quad (35)$$

2. We have a non-scattering, non-multiplying region, so the contribution from scattering and fission are removed.

$$\phi(\vec{r}, E) = \int_V dV' \frac{\exp(-\tau)}{4\pi|\vec{r}' - \vec{r}|^2} \int_0^\infty dE' S(\vec{r}', E) \quad (36)$$

3. Substitute the expression for $S(\vec{r}', E)$ and integrate with respect to E' .

$$= \int_V dV' \frac{\exp(-\tau)}{4\pi|\vec{r}' - \vec{r}|^2} \int_0^\infty dE' S(E) \delta(\vec{r}) \quad (37)$$

$$= \int_V dV' \frac{\exp(-\tau)}{4\pi|\vec{r}' - \vec{r}|^2} S(E) \delta(\vec{r}) \quad (38)$$

4. We are given the following expression for the optical thickness, τ .

$$\tau(E; \vec{r} - \rho' \hat{\Omega} \rightarrow \vec{r}) \equiv \int_0^{\rho'} d\rho'' \Sigma_t(\vec{r} - \rho'' \hat{\Omega}, E) \quad (39)$$

5. We know $\Sigma_t(E) = \Sigma_t$, which is constant. With this, τ becomes.

$$\tau(E; \vec{r} - \rho' \hat{\Omega} \rightarrow \vec{r}) \equiv \int_0^{\rho'} d\rho'' \Sigma_t \quad (40)$$

$$= \Sigma_t \int_0^{\rho'} d\rho'' \quad (41)$$

$$\tau(E; \vec{r} - \rho' \hat{\Omega} \rightarrow \vec{r}) = \Sigma_t \rho' \quad (42)$$

6. Where, ρ' represents the displacement from the source location $\vec{r}$.

$$\rho' = |\vec{r}' - \vec{r}| \quad (43)$$

$$\tau = \Sigma_t |\vec{r}' - \vec{r}| \quad (44)$$

7. Substituting the above expression back into the integral transport equation.

$$= \int_V dV' \frac{\exp(-\Sigma_t |\vec{\mathbf{r}}' - \vec{\mathbf{r}}|)}{4\pi |\vec{\mathbf{r}}' - \vec{\mathbf{r}}|^2} S(E) \delta(\vec{\mathbf{r}}) \quad (45)$$

$$= \frac{S(E)}{4\pi} \int_V dV' \frac{\exp(-\Sigma_t |\vec{\mathbf{r}}' - \vec{\mathbf{r}}|)}{|\vec{\mathbf{r}}' - \vec{\mathbf{r}}|^2} \delta(\vec{\mathbf{r}}) \quad (46)$$

8. With the integration over all space and the 3D, spatial, delta function at $\vec{\mathbf{r}}$, we can write the below expression. I am omitting the introduction of a temporary variable such as $\vec{\mathbf{r}}''$ while integrating over all space. Technically, I should have introduced the temporary variable to integrate over and not had the exact same variables in the integrand as I do after the integration, but this step is implied for brevity.

$$= \frac{S(E)}{4\pi} \frac{\exp(-\Sigma_t |\vec{\mathbf{r}}' - \vec{\mathbf{r}}|)}{|\vec{\mathbf{r}}' - \vec{\mathbf{r}}|^2} \quad (47)$$

9. We can write the final expression for scalar flux as:

$$\phi(\vec{\mathbf{r}}, E) = \frac{S(E) \exp(-\Sigma_t |\vec{\mathbf{r}}' - \vec{\mathbf{r}}|)}{4\pi |\vec{\mathbf{r}}' - \vec{\mathbf{r}}|^2} \quad (48)$$